Example 14.2

AR(1) process $\quad X_t = \phi X_{t-1} + Z_t$

$$\gamma(h) = \frac{\sigma_Z^2}{1-\phi^2}\phi^{|h|}$$

$$g(\omega) = \frac{1}{2\pi}\sum_{h=-\infty}^{\infty} \gamma(h) e^{i\omega h}$$

$$= \frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\left[\frac{\sigma_Z^2}{1-\phi^2}\phi^{|h|}\right] e^{i\omega h}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)}\sum_{h=-\infty}^{\infty}\phi^{|h|} e^{i\omega h}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)}\cancel{\sum}\left\{1 + \sum_{h=1}^{\infty}\phi^{|h|}\left(e^{i\omega h} + e^{-i\omega h}\right)\right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)}\left\{1 + 2\sum_{h=1}^{\infty}\phi^{|h|}\cos(\omega h)\right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)}\left\{1 + \sum_{h=1}^{\infty}\phi^{|h|}e^{i\omega h} + \sum_{h=1}^{\infty}\phi^{|h|}e^{-i\omega h}\right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)}\left\{1 + \underbrace{\sum_{h=1}^{\infty}\left(\phi e^{i\omega}\right)^h}_{\text{sum of a GP}} + \underbrace{\sum_{h=1}^{\infty}\left(\phi e^{-i\omega}\right)^h}_{\text{sum of a GP}}\right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)} \left\{ 1 + \frac{\phi e^{i\omega}}{1-\phi e^{i\omega}} + \frac{\phi e^{-i\omega}}{1-\phi e^{-i\omega}} \right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)} \left\{ 1 + \frac{\phi e^{i\omega}(1-\phi e^{-i\omega}) + \phi e^{-i\omega}(1-\phi e^{i\omega})}{(1-\phi e^{i\omega})(1-\phi e^{-i\omega})} \right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)} \left\{ 1 + \frac{\phi(e^{i\omega}+e^{-i\omega}) - 2\phi^2}{1+\phi^2-\phi(e^{i\omega}+e^{-i\omega})} \right\}$$

$$= \frac{\sigma_Z^2}{2\pi(1-\phi^2)} \left\{ 1 + \frac{2(\phi\cos\omega - \phi^2)}{1+\phi^2-2\phi\cos\omega} \right\}$$

$$= \cancel{\frac{\sigma_Z^2}{2\pi(1-\phi^2)}}$$

$$= \frac{\sigma_Z^2}{2\pi(1+\phi^2-2\phi\cos\omega)} \qquad \omega \in (-\pi, \pi]$$

Example 14.2 - New Approach

$$X_t = \phi X_{t-1} + Z_t$$

$$\Rightarrow (1 - \phi B) X_t = Z_t$$

$$\Rightarrow X_t = \frac{1}{(1 - \phi B)} Z_t$$

$$= \psi(B) Z_t \quad \text{where} \quad \psi(B) = \frac{1}{1 - \phi B}$$

$$g(\omega) = \frac{\sigma_Z^2}{2\pi} \psi(e^{i\omega}) \psi(e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi(1 - \phi e^{i\omega})(1 - \phi e^{-i\omega})}$$

$$= \frac{\sigma_Z^2}{2\pi(1 + \phi^2 - \phi(e^{i\omega} + e^{-i\omega}))}$$

$$= \frac{\sigma_Z^2}{2\pi(1 + \phi^2 - 2\phi \cos\omega)} \qquad \omega \in (-\pi, \pi]$$